

Cloud Computing

Virtual Resources

- A virtual resource is an abstraction
- The OS implements the abstraction through use of real resources
- An application can be migrated to any host offering the same abstraction

Virtualization and Cloud Computing

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Types of Virtualization

- Simulation and Emulation
- Full Virtualization
- Para-virtualization
- Hybrid Virtualization
- OS-Level Virtualization

Types of Virtualization: Simulation and Emulation

- Simulation
 - Simulators copy things from the real world into a virtual environment to give an idea about how that thing would work.
 - It simulates the basic behavior but doesn't necessarily follow all the rules of the real environment.
 - Running a model of some system to observe its behavior

Types of Virtualization: Simulation and Emulation

- Emulation
 - Emulation is the process of implementing the interface and functionality of one system on a system having a different interface and functionality
 - Emulation is basically a complete imitation of the real thing. It just operates in a virtual environment instead of the real world.
 - Emulation duplicates while simulation replicates a real device.

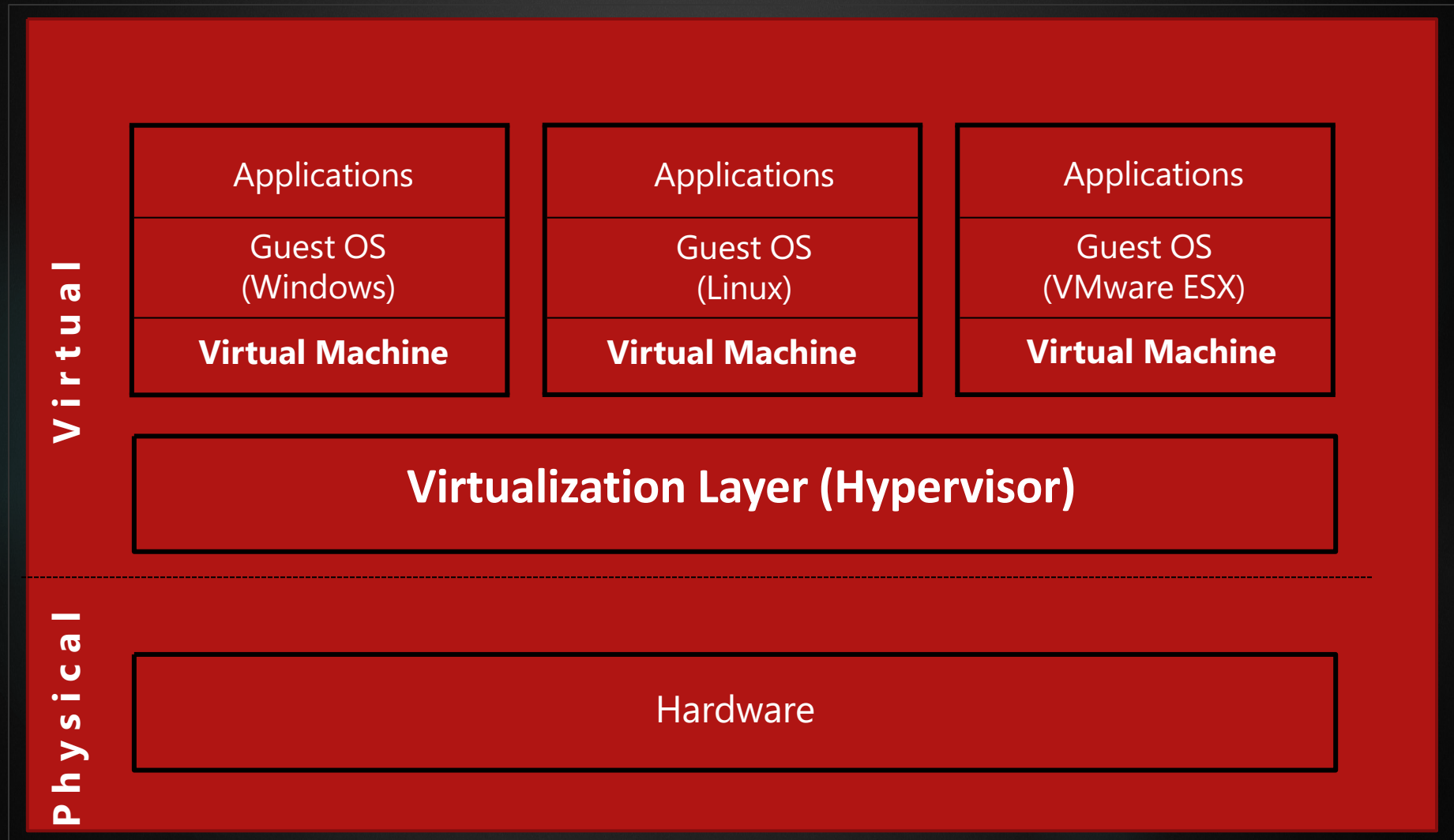
Types of Virtualization: Full Virtualization

- The functionally provides a complete set of hardware simulation for guest OS
- The virtual machine simulates hardware to allow an unmodified "guest" OS to be run in isolation.
- There is two type of Full virtualizations in the enterprise market.
 - Hardware-assisted full virtualization
 - Software assisted full virtualization

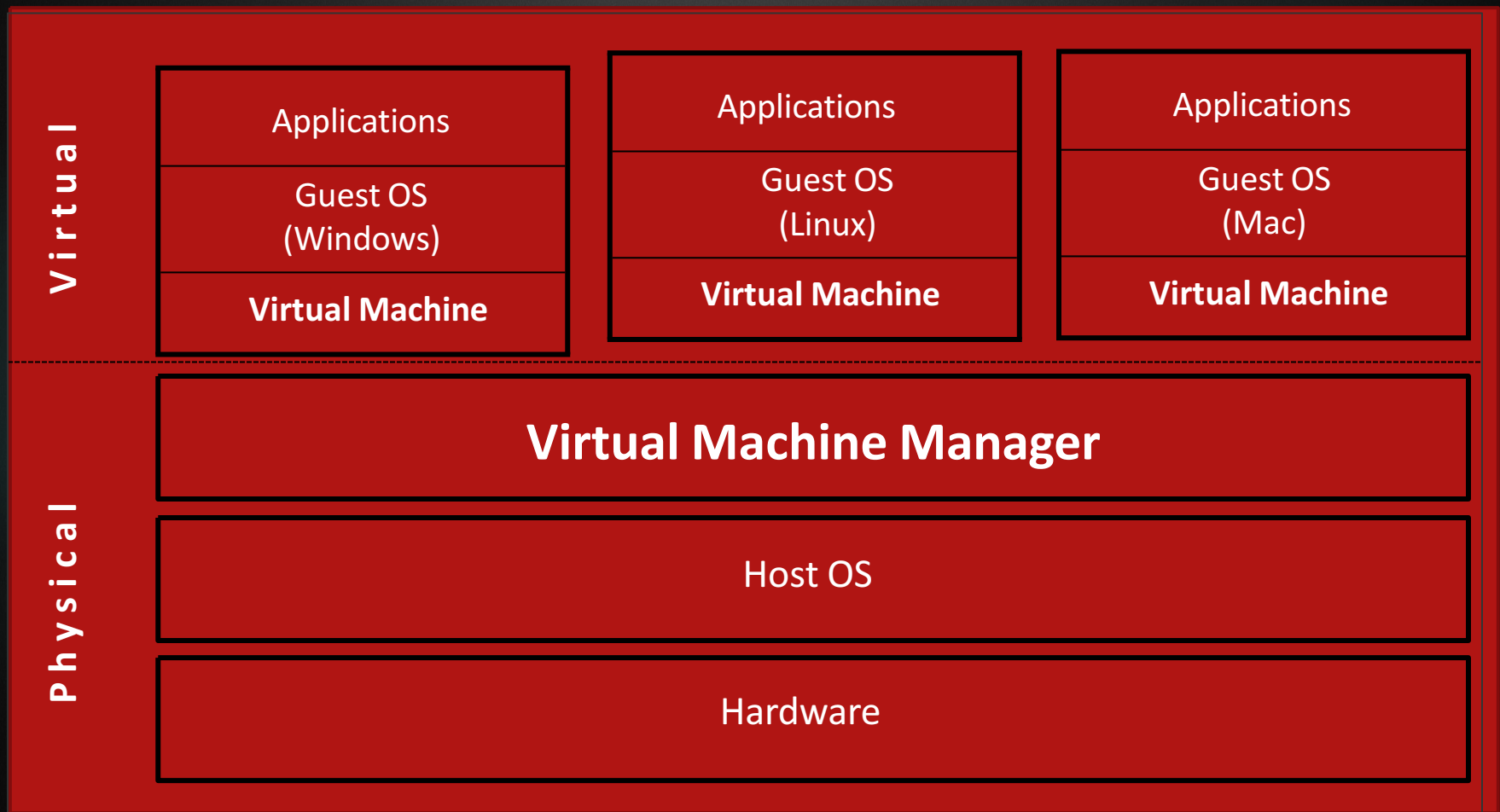
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Types of Virtualization: Hardware Assisted Full Virtualization



Types of Virtualization: Software Assisted Full Virtualization

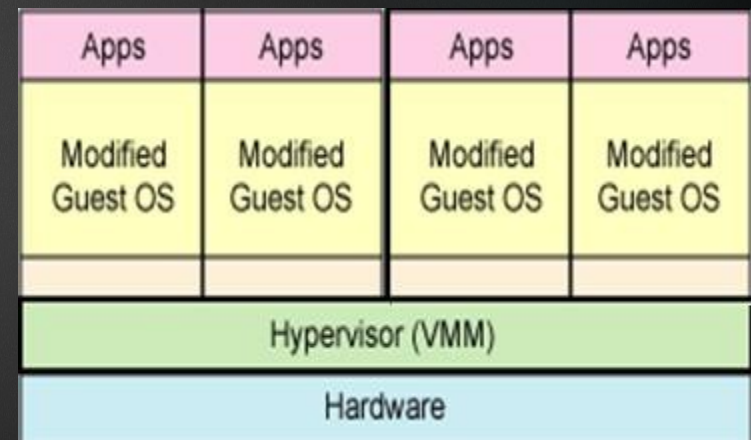


Types of Virtualization: Full Virtualization

- Types of Hypervisors
 - Type 1—Runs directly on computer's hardware
 - VMware ESXi /ESX
 - KVM
 - Hyper-V
 - Xen
 - Type 2—Runs as a process within existing operating system
 - VMware workstation
 - Virtual PC
 - VirtualBox

Types of Virtualization: Para-Virtualization

- It provides a virtual hardware abstraction that is similar, but not identical to the real hardware.
- Allow multiple guest OS to run and Xen hypervisor does the resource allocation.
- VM OS needs modification
- Examples:
 - Xen

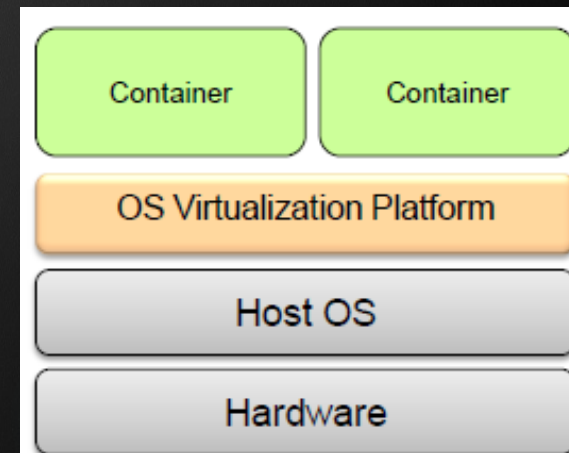


Types of Virtualization: Hybrid Virtualization

- In hardware assisted Full virtualization, Guest operating systems are unmodified and it involves many VM traps and thus high CPU overheads which limit the scalability.
- Para-virtualization is a complex method where guest kernel needs to be modified.
- For better performance, the virtual machine uses para-virtualization for specific hardware drivers and the host uses full virtualization for other features

Types of Virtualization: OS-Level Virtualization

- Virtualizing a physical server at the operating system level, enabling multiple isolated and secure virtualized servers to run on a single physical server.
- Virtualization platform on top of OS offers multiple containers
- Each container is a virtual OS (or Virtual Private Server)
- Examples
 - Linux-VServer
 - OpenVZ



Linux-Related Virtualization Projects

Project	Type	License
QEMU	Emulation	GPL
VMware	Full virtualization	Proprietary
Xen	Para-virtualization	GPL
Linux-VServer	Operating system- level virtualization	GPL